

MATH 344: Final Exam
Spring —₁ 2026
04/29/2026
150 Minutes

Name: _____

Write your name on the appropriate line on the exam cover sheet. This exam contains 16 pages (including this cover page) and 15 questions. Check that you have every page of the exam. Answer the questions in the spaces provided on the question sheets. Be sure to answer every part of each question and show all your work. If you run out of room for an answer, continue on the back of the page — being sure to indicate the problem number.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
8	10	
9	10	
10	10	
11	10	
12	10	
13	10	
14	10	
15	10	
Total:	150	

1. Let $\mathbf{u} = \langle \sqrt{3}, 1 \rangle$ and $\mathbf{v} = \langle \sqrt{3}, 3 \rangle$.

(a) (2 points) Compute $\mathbf{u} - 2\mathbf{v}$.

(b) (2 points) Are $\mathbf{u}$ and $\mathbf{v}$ perpendicular? Justify your answer.

(c) (5 points) Find the angle between $\mathbf{u}$ and $\mathbf{v}$.

(d) (1 point) Find a unit vector parallel to $\mathbf{u}$.

2. Consider the matrices A , B and the vector $\mathbf{u}$ given below.

$$A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 2 & 0 \\ 1 & 3 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 & -1 & 6 & 4 \\ 3 & 1 & 0 & 1 & 2 \\ 2 & 6 & 1 & -3 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}$$

(a) (3 points) Compute the $(2, 4)$ -entry of AB , $(AB)_{24}$, *without* explicitly computing AB entirely.

(b) (3 points) Compute $A\mathbf{u}$ by treating the product as a linear combination of columns of A .

(c) (4 points) Compute $\det A$.

3. Suppose matrices A, B both had the following row operations performed on it:

$$\begin{aligned}5R_1 &\rightarrow R_1 \\ R_2 &\longleftrightarrow R_3 \\ R_1 + R_2 &\rightarrow R_2 \\ R_1 - R_3 &\rightarrow R_3 \\ 2R_2 + R_3 &\rightarrow R_3\end{aligned}$$

After these row operations, the matrices A', B' below, respectively, were obtained:

$$A' = \begin{pmatrix} 10 & -2 & 4 \\ 0 & -1 & 6 \\ 0 & 0 & 3 \end{pmatrix} \quad B' = \begin{pmatrix} -3 & 0 & 1 & 4 & 0 \\ 0 & 1 & 5 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

(a) (3 points) Compute $\det A$.

(b) (2 points) Using (a), determine whether A is invertible, i.e. whether A^{-1} exists.

(c) (2 points) Determine the rank and nullity of B .

(d) (3 points) Does $B\mathbf{x} = \mathbf{0}$ have a solution? If so, is the solution unique?

4. (10 points) Matrices M, N, P below are in RREF and represent augmented matrices coming from systems of linear equations. For each matrix, determine whether the corresponding system is consistent or inconsistent. If it is inconsistent, explain why. If it is consistent, either find the unique solution or parametrize the solution set (if infinitely many solutions exist). Use variables $x_1, x_2, \dots$.

$$M = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

(a) System Corresponding to M :

(b) System Corresponding to N :

(c) System Corresponding to P :

5. (10 points) Write the matrix-vector equation corresponding to the system of equations below. Solve this system using the inverse of the coefficient matrix.

$$\begin{cases} 2x + 5y = 0 \\ x - y = -7 \end{cases}$$

6. (10 points) Find the LU decomposition of the matrix given below.

$$A = \begin{pmatrix} 3 & 2 & 1 \\ 9 & 5 & 8 \\ 0 & 2 & -9 \end{pmatrix}$$

7. (10 points) A matrix A can be written as $PA = LU$, where P, L, U are given below.

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & -4 \end{pmatrix}$$

Use this information to solve the equation $A\mathbf{x} = \begin{pmatrix} 5 \\ 0 \\ 3 \end{pmatrix}$.

8. (10 points) A matrix A and its reduced-row echelon form, R , are given below.

$$A = \begin{pmatrix} 4 & -8 & 7 & 4 \\ 1 & -2 & 2 & 1 \\ 3 & -6 & 5 & 3 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & -2 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(a) Determine which $\mathbb{R}^n$ each of the spaces below are a subspace of and find their dimension.

	$\text{Col}(A)$	$\text{Row}(A)$	$\text{Null}(A)$
Subspace of...			
Dimension			

(b) Find a basis for $\text{Col}(A)$.

(c) Find a basis for $\text{Row}(A)$.

(d) Find a basis for $\text{Null}(A)$.

9. (10 points) Mark the statements below as True (T) or False (F).

- (a) _____: The set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly independent if and only if at least one of the vectors can be written as a linear combination of the other vectors.
- (b) _____: The set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly independent if and only if the equation $c_1\mathbf{v}_1 + \dots + c_n\mathbf{v}_n = \mathbf{0}$ has only the trivial solution.
- (c) _____: If $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is a set of vectors in $\mathbb{R}^n$ with $k > n$, then the set is linearly dependent.
- (d) _____: The columns of *any* 6×7 matrix are linearly dependent.
- (e) _____: The set $\{\mathbf{u}, \mathbf{v}\}$ is linearly dependent if and only if $\mathbf{u}$ and $\mathbf{v}$ are parallel.

10. (10 points) For each of the following, indicate the letter of the correct response.

- (i) ____: If the null space of a 9×5 matrix A is 3-dimensional, which of the following is the dimension of the column space of A ?
- (a) 2
 - (b) 3
 - (c) 5
 - (d) 9
 - (e) Cannot be determined, more information is required.
- (ii) ____: If A is a 6×7 matrix with rank 6, then how many solutions does $A\mathbf{x} = \mathbf{b}$ have for $\mathbf{b} \in \mathbb{R}^7$?
- (a) None
 - (b) One
 - (c) Two
 - (d) Infinitely Many
 - (e) More information required
- (iii) ____: If A is a 3×4 matrix, then what is the largest the rank of A can be?
- (a) 0
 - (b) 1
 - (c) 3
 - (d) 4
 - (e) Cannot be determined.
- (iv) ____: If A is a 3×5 matrix, what is the smallest the nullity of A can be?
- (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
 - (e) 5
- (v) ____: If A is a 4×4 matrix, which of the following would imply that A is invertible?
- (a) $\text{rank } A + \text{nullity } A = 4$
 - (b) $\text{rank } A = 4$
 - (c) $\text{nullity } A = 0$
 - (d) $\text{nullity } A > 0$
 - (e) (b) *and* (c)

11. (10 points) Find the eigenvalues, eigenvectors, and all associated eigenspaces for the matrix $A = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$.

12. (10 points) Let A, B, C be matrices. The characteristic polynomials, eigenvalues, and a basis for the associated eigenspaces for each of A, B, C are given below. Only one of A, B, C is diagonalizable. Identify which matrix is diagonalizable and find matrices P, D such that the given matrix is equal to PDP^{-1} .

Matrix	Characteristic Polynomial	Eigenvalue	Eigenspace Basis
A	$p(\lambda) = (\lambda - 1)^2$	1	$\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$
B	$p(\lambda) = -(\lambda + 2)^2(\lambda - 1)$	-2	$\left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 8 \\ -1 \end{pmatrix} \right\}$
		1	$\left\{ \begin{pmatrix} 1 \\ -3 \\ 3 \end{pmatrix} \right\}$
C	$p(\lambda) = (\lambda - 1)^2(\lambda - 5)^2$	1	$\left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right\}$
		5	$\left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$

13. (10 points) Find an orthonormal basis for the subspace of $\mathbb{R}^3$ generated by the span of the vectors below.

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}$$

14. (10 points) Find the QR factorization for the matrix $A = \begin{pmatrix} 3 & 1 \\ 4 & 1 \end{pmatrix}$.

15. (10 points) Find the line of best fit (the least square regression line) for the dataset given below.

$$\begin{array}{c|ccc} x & -1 & 0 & 1 \\ \hline y & -1 & 1 & 1 \end{array}$$